

Example: /functional-unit polyester_tshirt

This is a recorded example of the functional-unit teaching skill in action, using the polyester_tshirt case study from skills_references/polyester_tshirt/.

The skill reads recipe_card.md for that case study and teaches the concept of functional unit using real data, Socratic questions, and fashion business context.

The conversation

AI: Imagine you're a sustainability manager at a fashion brand, and two of your T-shirt suppliers both hand you a report claiming their product has a lower carbon footprint than the other.

What is the very first question you would ask before accepting either of those claims?

Student: How was it measured?

AI: That's exactly the right instinct — and “how was it measured?” is the question that separates a sustainability professional from someone who just reads the headline number.

The most important part of “how was it measured?” is actually: **measured per what?** Per shirt? Per kilogram of fabric? Per year of wearing? A T-shirt that looks better per kilogram of material might look worse per finished garment if it uses more fabric. The number is meaningless without knowing what it's attached to.

In LCA — Life Cycle Assessment, the method used to calculate environmental impact — this is called the **functional unit**. It's the precise definition of exactly what is being measured, so that two products can be compared on equal footing.

For this polyester T-shirt study, the functional unit is: **1 polyester T-shirt, ready for sale (approximately 200 g of fabric)**.

Not per kilogram of polyester, not per year of wearing — per finished shirt at the point it leaves the factory. That unit was chosen because it matches what a buyer actually purchases and what a brand actually sells.

Here's a question to test that idea: what if one brand reports their footprint **per shirt** and a competitor reports theirs **per kilogram of fabric** — and the competitor's shirt happens to be heavier? Could the competitor make their shirt look better than it really is just by choosing a different unit?

Student: Yes, but I'm not exactly sure how.

AI: That's a perfectly natural place to be — the intuition is right, the mechanics just need a moment to click.

Here's a concrete example. Say Brand A's shirt weighs 200 g and has a footprint of **2.58 kg CO₂ per shirt**. Brand B's shirt weighs 300 g and has a footprint of **3.87 kg CO₂ per shirt** — actually worse in total. But if Brand B reports **per kilogram of fabric** instead, their number becomes $3.87 \div 0.3 = 12.9 \text{ kg CO}_2 \text{ per kg}$, while Brand A's becomes $2.58 \div 0.2 = 12.9 \text{ kg CO}_2 \text{ per kg}$ too — they look identical. Brand B hid the fact that their shirt uses 50% more material just by switching the unit.

Flip it the other way and a heavier shirt can actually look *better* per kilogram, because the fixed costs of running the factory get spread over more fabric. The unit choice isn't just a technicality — it can completely reverse a comparison.

This is why sustainability standards like the **Higg Index** (used widely in fashion) require the functional unit to be declared upfront in any environmental claim. Without it, the number can't be verified or compared — it's just a figure floating in the air.

As a buyer or merchandiser, the habit to build is simple: whenever a supplier hands you a sustainability number, the first question is always **"per what?"** — and if they can't answer clearly, the number isn't ready to act on.

Student: So what are some other examples of functional units for fashion business or retail clothing stores?

AI: That's a great question to sit with — and it shows you're already thinking like someone who would design these studies, not just read them.

Here are some real examples from fashion and retail, each chosen for a different reason:

- **Per garment** — the most common for finished products like T-shirts, jeans, or jackets. Matches what a customer buys and what a brand sells. Easy to communicate on a hangtag.
- **Per kilogram of fibre** — used when comparing raw materials like wool vs cotton vs polyester at the supplier level, before anyone has cut or sewn anything.
- **Per wear** — divides the total footprint by how many times a garment is worn over its lifetime. Favours durable, expensive items over cheap fast fashion.

Used by brands like Patagonia to argue that a £300 jacket is actually lower impact than a £30 one.

- **Per square metre of fabric** — common in home textiles (curtains, upholstery) where the product is sold by area rather than by piece.
- **Per season's wardrobe** — occasionally used in academic research to compare the total impact of buying fast fashion vs buying less but better quality over a full year.

The choice always comes back to the same question: **what decision is this number supposed to help someone make?** A sourcing manager comparing fibre suppliers needs per kilogram. A customer choosing between two jackets needs per wear. A brand reporting to investors might use per garment.

Would you like to try the same concept with a different product — or shall we move on to the next part of the T-shirt study?

Student: Do you have a graph that shows this example?

AI: We have two supply chain diagrams for the polyester T-shirt — but they show the structure of the supply chain (the three process boxes and arrows), not a comparison between functional units. Here is the structure diagram:

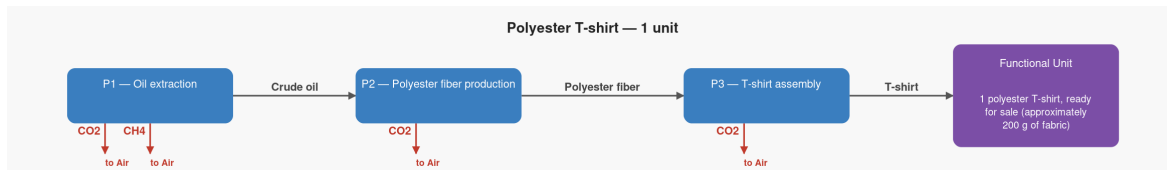


Figure 1: Polyester T-shirt supply chain — structure

The functional unit lives at the **rightmost box — P3, T-shirt assembly**. That is the reference process: the step that delivers exactly one finished T-shirt, ready for sale. Everything to its left (the fiber factory, the oil well) exists only to supply it. When the study says “1 polyester T-shirt, ready for sale,” it is pointing at the output of that box — the moment the shirt is complete and the measurement stops.

A functional unit *comparison* chart — something that visually shows how the same product looks different depending on which unit you choose — doesn't exist yet in the reference materials, but it would be a useful addition.

Want to keep going with the T-shirt study and look at the supply chain in more detail, or is there something else you'd like to explore?

What this example demonstrates

- The opening question was framed as a real professional scenario (sustainability manager, two competing supplier claims) rather than an abstract definition — the student arrived at “how was it measured?” immediately, which is the right instinct
- The skill deepened “how was it measured?” into “per what?” before revealing the functional unit, so the concept landed with concrete motivation behind it
- The what-if example used real numbers from the recipe card (2.58 kg CO₂, 200 g shirt) alongside a invented Brand B scenario — keeping the teaching grounded in the actual case study
- The student’s follow-up question about other functional unit examples (not in the skill script) was answered naturally using fashion-specific contexts: per wear, per kg of fibre, per square metre, per season
- When the student asked for a graph, the skill was honest that the available diagrams show supply chain structure, not functional unit comparisons — then used the structure SVG to locate the functional unit within the diagram (the reference process box, P3) rather than deflecting the question
- Every technical term introduced (LCA, functional unit, Higg Index, reference process) was explained immediately in plain language without interrupting the flow of conversation